

IAT-882 Assignment 3

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A1

1. Dropped the `geom` column because it is all empty.
2. Removed rows where `geometry` data isn't there. Since `geometry` and `geo\_point\_2d` has same data, we removed `geometry` and keep `geo\_point\_2d`.
3. Filtered down only to where `status` is `Issued`. We are only concerned with "active" businesses. It's a reasonable filter down to a smaller, more manageable size.
4. If `feepaid` is missing, just input 0 there. We will use that column to determine size of company, and a reasonable compromise is to treat it as "fee not paid"
5. 46.62% of the `postalcode` column is NaN. I won't modify it because for my later analysis in Part B, I will use `localarea`, that only has 1% NaN, which I will be comfortable dropping then.
6. For business type, we simplified and now have these:
 - Financial
 - Healthcare
 - Real Estate and Housing
 - Food and Drink
 - Service Provider
 - Art and Entertainment
 - Adult
 - Retail Dealer
 - Other
7. Removed rows where there isn't data on licence dates

A2

1. The K-means clustering roughly reflects the key neighborhoods of Vancouver. For example:
 - UBC pointgrey
 - Marpole
 - Downtown
 - Kitsilano
 - Oakridge
2. The DBSCAN on the other hand doesn't segment as nicely into neighborhoods. Since it doesn't care about shape, the clustering doesn't map into our idea of Vancouver neighborhoods. Furthermore, a lot of locations are classified as noise. The noise are just locations that don't have min_sample locations within it's radius.

A3

This is much more spread apart on a geographical map. I would expect more adult businesses to be near downtown, but from a size perspective, they are not as tightly knit as the lat-lon clusters as expected. The v-shaped graph also shows very diverging patterns.

B4

There are neighborhoods that fall in the same cluster despite being far apart. And of course, how similar the neighborhoods are to each other depends on how many clusters we select. The more clusters we select, the more fine-grained the grouping.

For example, with 7 clusters:

- Most are unique
- Kensington, Renfrew, Victoria Frasersview, Riley Park are similar - that makes sense, these are slightly older, more residential areas

For example, with 3 clusters:

- Downtown, WestEnd, and Kitsilano are similar, more vibrant areas for younger millennials
- Mount pleasant and Hastings are similar, which is slightly surprising, but also makes sense, with more industrial type eateries and breweries, etc.

Downtown has most businesses, largest point, which makes sense